

ROS2 1일차 과제1 실행 결과 보고서

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1. 실행

결과

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minchan@minchan-IdeaPad-Slim-3-15ABR8: ~$ ros2 run my_cpp_pkg my_publisher
[INFO] [1789464477.803667487] [my_pub]: Publishing: id=0, label='hello'
[INFO] [1789464478.303756572] [my_pub]: Publishing: id=1, label='hello'
[INFO] [1789464478.803889184] [my_pub]: Publishing: id=2, label='hello'
[INFO] [1789464479.303989971] [my_pub]: Publishing: id=3, label='hello'
[INFO] [1789464479.803948573] [my_pub]: Publishing: id=4, label='hello'
[INFO] [1789464480.303629712] [my_pub]: Publishing: id=5, label='hello'
[INFO] [1789464480.803622489] [my_pub]: Publishing: id=6, label='hello'
^C[INFO] [1789464481.064450947] [rclcpp]: signal_handler(SIGINT/SIGTERM)
minchan@minchan-IdeaPad-Slim-3-15ABR8: ~$

minchan@minchan-IdeaPad-Slim-3-15ABR8: ~$ ros2 run my_cpp_pkg my_subscriber
[INFO] [1789464477.803984121] [my_sub]: I heard: id=0, label='hello'
[INFO] [1789464478.304027534] [my_sub]: I heard: id=1, label='hello'
[INFO] [1789464478.803729288] [my_sub]: I heard: id=2, label='hello'
[INFO] [1789464479.303978309] [my_sub]: I heard: id=3, label='hello'
[INFO] [1789464479.803948066] [my_sub]: I heard: id=4, label='hello'
[INFO] [1789464480.303911499] [my_sub]: I heard: id=5, label='hello'
[INFO] [1789464480.803912027] [my_sub]: I heard: id=6, label='hello'
[INFO] [1789464605.857658509] [my_sub]: I heard: id=0, label='hello'
[INFO] [1789464606.357730826] [my_sub]: I heard: id=1, label='hello'
[INFO] [1789464606.857658492] [my_sub]: I heard: id=2, label='hello'
[INFO] [1789464607.358121068] [my_sub]: I heard: id=3, label='hello'
[INFO] [1789464607.858020491] [my_sub]: I heard: id=4, label='hello'
[INFO] [1789464608.358054599] [my_sub]: I heard: id=5, label='hello'
[INFO] [1789464608.857998751] [my_sub]: I heard: id=6, label='hello'

minchan@minchan-IdeaPad-Slim-3-15ABR8: ~$ cd ~/ros2_ws/src
minchan@minchan-IdeaPad-Slim-3-15ABR8: ~/ros2_ws/src$ ros2 bag record /my_topic
[WARN] [ros2bag]: Positional "topics" argument deprecated. Please use optional "--topics" argument instead.
[INFO] [1789464447.338495097] [rosbag2_recorder]: Press SPACE for pausing/resuming
[INFO] [1789464447.338513184] [rosbag2_recorder]: Event publisher thread: Started
[INFO] [1789464447.342542662] [rosbag2_recorder]: Starting recording to 'rosbag2_2026_09_15-18_27_27'
[INFO] [1789464447.344789952] [rosbag2_recorder]: Listening for topics...
[INFO] [1789464447.344814813] [rosbag2_recorder]: Recording...
[INFO] [1789464447.344829548] [rosbag2_recorder]: Topics discovery started.
[INFO] [1789464477.310837734] [rosbag2_recorder]: Subscribed to topic '/my_topic'
[INFO] [1789464477.310872023] [rosbag2_recorder]: All requested topics are subscribed. Stopping discovery...
[INFO] [1789464477.310885432] [rosbag2_recorder]: Topics discovery stopped.
[INFO] [1789464504.995634227] [rosbag2_recorder]: Pausing recording.
[INFO] [1789464504.995766357] [rosbag2_cpp]: Writing remaining messages from cache to the bag. It may take a while
[INFO] [1789464504.998952620] [rosbag2_recorder]: Recording stopped
[INFO] [1789464505.000531680] [rosbag2_recorder]: Event publisher thread: Exited
minchan@minchan-IdeaPad-Slim-3-15ABR8: ~/ros2_ws/src$

minchan@minchan-IdeaPad-Slim-3-15ABR8: ~$ cd ~/ros2_ws/src
minchan@minchan-IdeaPad-Slim-3-15ABR8: ~/ros2_ws/src$ ros2 bag play rosbag2_2026_09_15-18_27_27
[INFO] [1789464605.740483880] [rosbag2_player]: Set rate to 1
[INFO] [1789464605.754627091] [rosbag2_player]: Adding keyboard callbacks.
[INFO] [1789464605.754669125] [rosbag2_player]: Press SPACE for Pause/Resume
[INFO] [1789464605.754684558] [rosbag2_player]: Press CURSOR_RIGHT for Play Next Message
[INFO] [1789464605.754697758] [rosbag2_player]: Press CURSOR_UP for Increase Rate 10%
[INFO] [1789464605.754708792] [rosbag2_player]: Press CURSOR_DOWN for Decrease Rate 10%
[INFO] [1789464605.755954122] [rosbag2_player]: Playback until timestamp: -1
minchan@minchan-IdeaPad-Slim-3-15ABR8: ~/ros2_ws/src$
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2. 작동 시나리오

*코드는 과제1의 publisher와 subscriber 재사용

rosbag을 이용한 토픽 기록 및 재생

publisher와 subscriber가 서로 통신하는 상태에서, 별도의 터미널에서 "/my_topic" 토픽을 기록하는 명령을 실행하여 해당 토픽에 발행되는 모든 메시지가 실시간으로 파일에 저장된다. 기록을 중단하면 파일 저장이 종료된다.

이렇게 저장된 기록 파일을 재생 명령을 통해 다시 불러와 재생하면 기록 당시 publisher가 발행했던 메시지들이 저장된 순서와 시점 그대로 다시 토픽으로 발행된다. 이미 실행 중인 subscriber는 실제 퍼블리셔가 다시 켜지지 않은 상태에서도, 재생되는 메시지를 정상적으로 수신하여 이전과 동일한 id와 label 값을 로그로 출력한다.